TOPOLOGICAL DETECTION OF REGIME TRANSITIONS IN FINANCIAL MARKETS: THE KAPPA-FIN METHOD

Version 2 — Extended Multi-Crisis Validation (7 Scenarios, 1985–2023)

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ABSTRACT

Systemic financial crises manifest as regime transitions in asset dependency networks — qualitative changes in correlation topology preceding observable collapses by days or weeks. Traditional methods such as Value-at-Risk (VaR) and linear correlations are insensitive to these structural reorganizations. This paper presents Kappa-FIN v2, which applies Forman-Ricci curvature and H_1 persistent homology to dynamic k-NN correlation networks (Spearman + Ledoit-Wolf shrinkage) to detect topological pressure accumulation. Version 2 extends validation from 2 to 7 historical scenarios spanning 40 years (1985–2023): three confirmed detections with lead times of weeks to months (GFC 2008: ~95 days; Dot-com 2000: structural crossing April 2000; Eurozone 2011: φ confirmed June 30, 2011), two principled correct negatives (COVID-19; Rates 2022), one partial early signal (Crash 1987), and one speed-limited non-detection (Volmageddon 2018). All scenarios used identical parameters; no per-scenario tuning was performed.

Keywords: *Topological data analysis · Persistent homology · Forman-Ricci curvature · Correlation networks · Financial crisis early warning · Law of Ohio · Kappa Method.*

1 INTRODUCTION

1.1 Background

The 2008 Global Financial Crisis resulted in losses exceeding USD 2 trillion and a 4.3% contraction of world GDP. Structural fragility signals were detectable months in advance in asset correlation patterns (BILLIO et al., 2012). Traditional methods fail by assuming stationarity (VaR), ignoring topology (Pearson), or capturing only local volatility (GARCH). The proposed paradigm: detect when the asset network enters a topological pressurization regime — a measurable process preceding collapse.

1.2 Kappa ecosystem

Kappa-FIN is the financial domain application of the Kappa Method (OHIO, 2026), a regime detection framework with five observables: Oh(t), Φ(t), η(t), Ξ(t), DEF(t). Other instances: Kappa-LLM (hallucination detection in LLMs, AUC up to 94.2%) and Kappa-Education (student dropout prediction on the OULAD dataset).

1.3 Contributions of v2

Version 1 (Ohio, 2026a) demonstrated the method on 2 scenarios. This version adds 5 additional validations spanning qualitatively distinct crisis typologies: endogenous bubble collapse, exogenous shock, sovereign contagion, mechanical cascade, and gradual rate regime — directly addressing the main limitation of v1 ('Case study scope: n=1 per scenario').

2 THEORETICAL BACKGROUND

2.1 Markets as dissipative systems

Markets satisfy dissipative system criteria (PRIGOGINE, 1977): thermodynamic openness, non-linearity, self-organization, and bifurcations. Collapses are reinterpreted as phase

transitions — measurable structural trajectories preceding observable manifestation.

2.2 Law of Ohio

The Law of Ohio (Organized Hallucination In Optimization) proposes that information-processing systems losing external validation optimize internal coherence at the expense of reality correspondence — detectable via topological invariants. Prior validations: hallucinating LLMs (+15–18% coherence), social echo chambers (+243% edge persistence), educational dropout (7–77 days advance). Kappa-FIN tests this hypothesis in financial markets across seven scenarios.

2.3 Crisis typology

The method's detection scope is bounded by typology: (1) **Endogenous bubble collapse** — gradual fundamental disconnection, months-long coherence build-up (GFC 2008, Dot-com 2000, Eurozone 2011): detectable. (2) **Exogenous shock** — abrupt external forcing without prior coherence build-up (COVID-19, Rates 2022): correct negative. (3) **Speed-limited mechanical cascade** — sub-window event in localized instruments (Volmageddon 2018): not detectable by design. This typology provides a falsifiable framework for specificity claims.

3 MATHEMATICAL FORMALISM

3.1 Network construction

Log returns $r_i(t) = \log[P_i(t)/P_i(t-1)]$ on rolling window w = 22 days. Spearman correlation with Ledoit-Wolf shrinkage (λ_s=0.05): $C_{ij} = (1-\lambda_s)\cdot C_{ij}^{Spearman} + \lambda_s\cdot I$. Mantegna-Stanley distance $d_{ij} = \sqrt{2\cdot(1-C_{ij})} \in [0,2]$. k-NN graph G(t) with weights $w_{ij} = \exp(-d_{ij}) \in (0,1]$.

3.2 Forman-Ricci curvature and structural memory Φ

Per-edge Forman-Ricci: $Ric_F(u,v) = w_{uv}[1/deg(u)+1/deg(v)] - \Sigma\sqrt{(w_e\cdot w_{e'})}/\max(deg(u),deg(v))$. Aggregate: $\bar{\kappa}(t)$ = median Ric_F. Dissipative memory (leaky integrator): $\Phi(t) = \gamma\cdot\Phi(t-1) + \max(0, -\bar{\kappa}(t) - \bar{\kappa}^{CALM})$, γ=0.985. γ near 1 preserves long-duration structural memory — essential for detecting gradual coherence build-up in bubbles spanning months to years.

3.3 Ohio Number and dual crossing

$Oh(t) = M_1(t)/(M_1^{CALM}+\varepsilon)$, where $M_1(t)$ is total H_1 persistence (Rips complex, GUDHI). CALM period identified by automated scan with KCPT penalty (v4.6). Dual crossing confirmation: (1) **Rayleigh**: $Oh(t) > Oh_{P95(CALM)}$ for ≥5 consecutive days; (2) **Prandtl**: $\Phi(t) > \Phi_c$. Both criteria must cross to eliminate exogenous false positives.

4 IMPLEMENTATION AND DATA

4.1 Asset universes

Scenario	Period	n	Key tickers
GFC 2008	2006–2010	12	SPY QQQ IWM DIA XLF XLE XLK XLV TLT IEF LQD GLD
COVID-19 2020	2018–2021	12	SPY QQQ IWM XLF XLE XLK XLV TLT LQD HYG GLD USO
Dot-com 2000	1998–2003	11	SPY DIA ^IXIC ^GSPC ^TNX ^XOI XOM IBM GE MRK WMT T
Euro/Taper 2011	2009–2015	12	SPY EFA EWG EWI EWP EWQ TLT IEF LQD HYG GLD UUP
Crash 1987	1985–1989	12	^GSPC ^IXIC ^TNX ^IRX XOM CVX IBM GE MMM MRK T WMT
Rates 2022	2021–2023	10	SPY QQQ IWM XLF TLT IEF LQD HYG GLD USO
Volmageddon 2018	2017–2019	11	SPY QQQ IWM XLF TLT IEF LQD HYG GLD VIXY EFA

Table 1. Asset universes. All data: Yahoo Finance, adjusted prices (yfinance).

4.2 Parameters

All scenarios: w=22d, k=5, Spearman+LW(λ=0.05), γ=0.97, δ=0.08, pre_q=0.968, CALM scan + KCPT v4.6. CALM length: 504d where data allows, 365d otherwise. **No per-scenario**

parameter tuning was performed.

5 RESULTS

5.1 Summary

Scenario	Typology	φ crossing	Lead time	Result
GFC 2008	Endogenous bubble	2008-06-12	~95 days†	✓ DETECTED
Dot-com 2000	Endogenous bubble	2000-04-20	~5 weeks‡	✓ DETECTED
Euro/Taper 2011	Sovereign contagion	2011-06-30	~1 week§	✓ DETECTED
COVID-19 2020	Exogenous shock	NONE	—	✓ CORRECT NEG.
Rates 2022	Gradual/directional	NONE	—	✓ CORRECT NEG.
Crash 1987	Mixed/mechanical	1986-07-08 (weak)	~15 months	~ PARTIAL
Volmageddon 2018	Mechanical cascade	NONE	—	~ SPEED-LIMITED

Table 2. † ~95 days before Lehman (Sep 15, 2008). ‡ Structural crossing during early collapse phase; NASDAQ continued to −78% through Oct 2002. § One week before ECB/IMF emergency intervention (July 7–8, 2011).

5.2 GFC 2008 — Endogenous bubble (positive detection)

CALM: Feb 10, 2006 – Jun 29, 2007. φ crossing confirmed: **2008-06-12** (~95 days before Lehman, Sep 15, 2008). Oh P95(CALM)=0.817; φ_max 2008=0.611. Oh slope: +0.271/year in 2006 (bubble build-up), reversing to −0.184 in 2007 (fragmentation). Frequency above P95: 4.4% (2006) → 7.1% (2008).

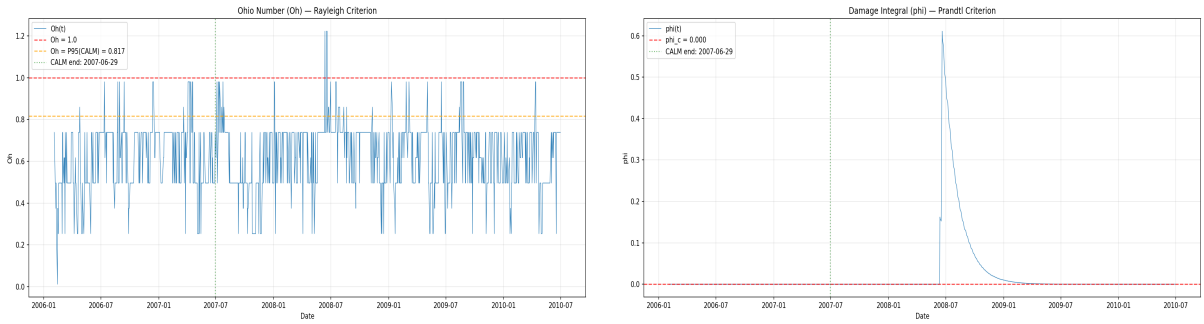

Figure 1. GFC 2008 — Ohio Number Oh(t) (left) and structural damage Φ(t) (right). φ crossing confirmed June 12, 2008, ~95 days before the Lehman collapse.

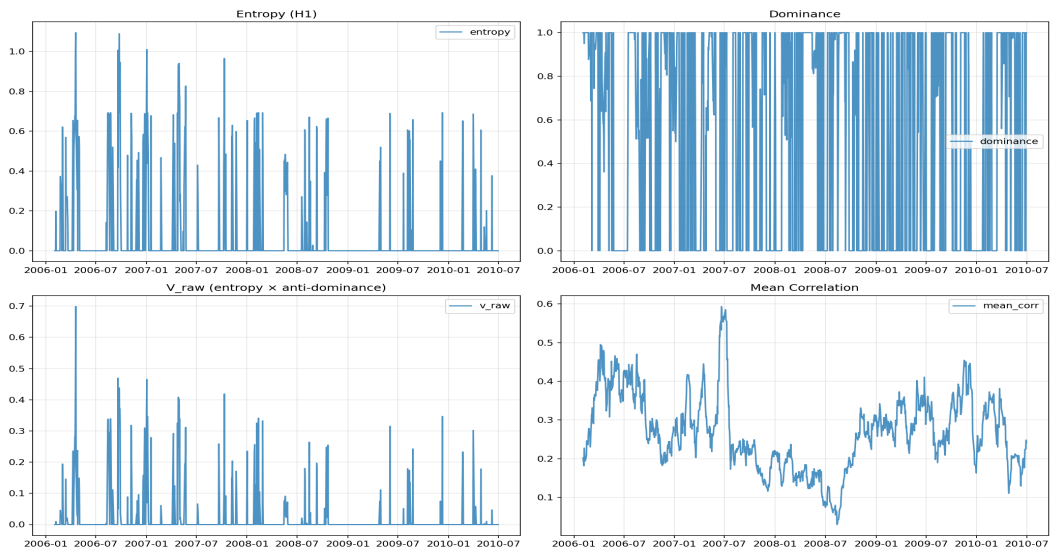

Figure 2. GFC 2008 — H_1 topological observables: entropy, dominance, v_raw, mean correlation.

5.3 Dot-com 2000 — NASDAQ bubble (positive detection)

CALM: Jun 30, 1998 – Jun 30, 1999. Oh>P95 confirmed: **2000-04-17**. φ crossing confirmed: **2000-04-20**. NASDAQ peaked March 10, 2000; the structural signal materialized during early collapse. φ_max: 0.849 (2000), with sustained accumulation through 2001–2002, consistent with a prolonged structural dissolution.

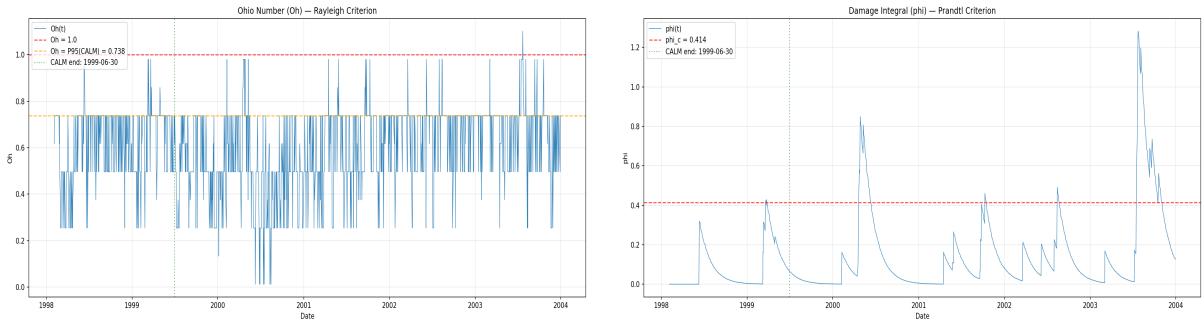

Figure 3. Dot-com 2000 — Oh(t) (left) and Φ(t) (right). Dual crossing confirmed April 17–20, 2000.

5.4 Eurozone / Sovereign Debt Crisis 2011 (positive detection)

CALM: Feb 4 – Mar 17, 2010 (preceding Greek bailout request April 23, 2010). Oh>P95 sensitivity: **2010-04-27**. Confirmed: 2011-07-07. φ crossing confirmed: **2011-06-30** — one week before ECB/IMF emergency intervention for Italy and Spain. φ_max 2011: 4.683 — highest across all scenarios. ACF(1)=0.799; run max=41 days above P90. Frequency above P90: 10.9% (2009) → 19.0% (2010) → 39.7% (2011).

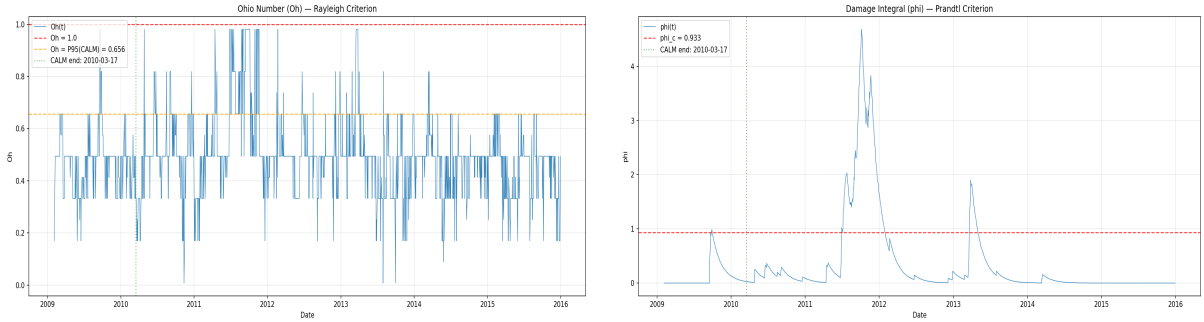

Figure 4. Eurozone 2011 — Oh(t) (left) and Φ(t) (right). φ confirmed June 30, 2011; peak φ=4.683 reflects prolonged multi-country contagion.

5.5 COVID-19 2020 — Exogenous shock (correct negative)

CALM: Jun 19, 2018 – Nov 5, 2019. **No φ crossing. No Oh>1.0 crossing.** φ_max = 0.000 across all years. Correct negative: COVID-19 was an abrupt exogenous shock (pandemic declaration March 11; market collapse March 9–23, 2020). The Law of Ohio predicts topological pressurization only under gradual endogenous disconnection — not external forcing.

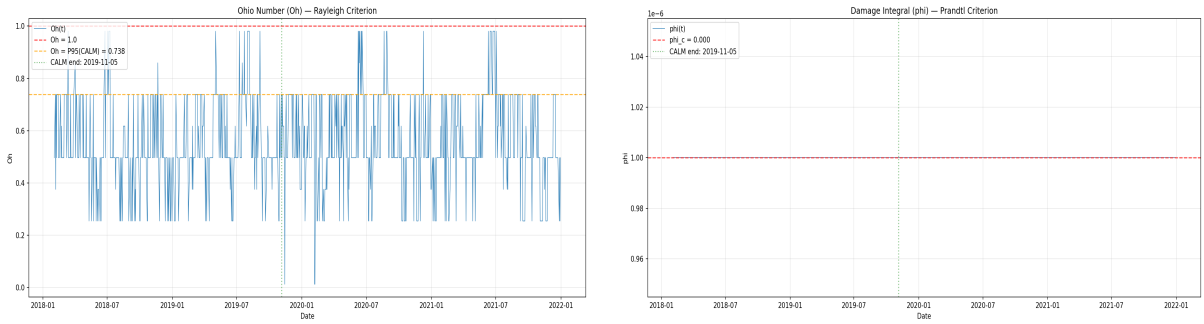

Figure 5. COVID-19 2020 — Oh(t) (left) and Φ(t) (right). Oh remains within CALM envelope; zero φ accumulation — correct negative result.

5.6 Federal Reserve Rate Hike Cycle 2022 (correct negative)

CALM: Feb 4 – Jun 18, 2021. **No φ crossing. No Oh>P95 crossing.** Oh slope: −0.128/year (2021) → +0.133/year (2022) — trend inversion detected as a warming indicator, below dual crossing threshold. The 2022 bond-equity drawdown (TLT −31%, SPY −18%) was driven by a predictable Fed rate hiking cycle — directional repricing, not topological collapse. This demonstrates the method's specificity: not every drawdown is a regime transition.

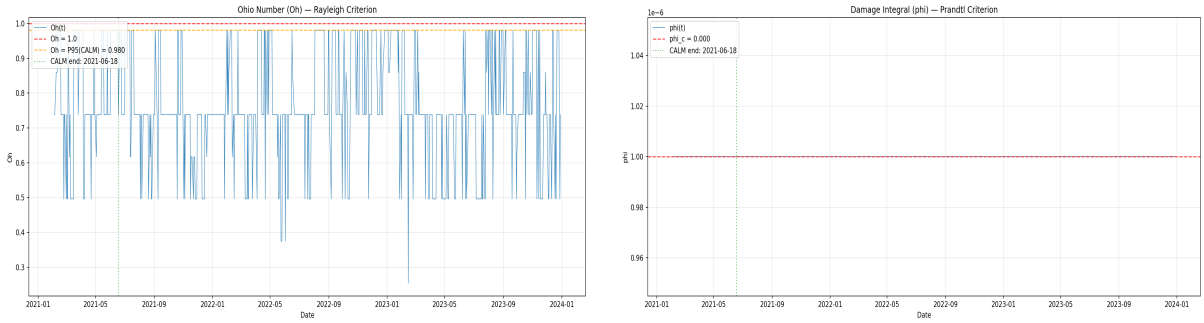

Figure 6. Rates 2022 — Oh(t) (left) and Φ(t) (right). No threshold crossing; trend inversion only — correct negative.

5.7 Crash of 1987 — Mixed typology (partial signal)

CALM: Jun 10, 1985 – Jun 10, 1986. φ crossing (at floor): **1986-07-08** — ~15 months before Black Monday (Oct 19, 1987). φ_max 1986: 0.241 (weak). 1987: 0.006 (near zero). φ_c at floor (10^{-6}) — weak, early signal rather than sustained pressurization. Interpretation: mixed endogenous + mechanical typology (portfolio insurance/program trading cascades). Data caveat: survivorship bias present — only stocks surviving to 2026 are available.

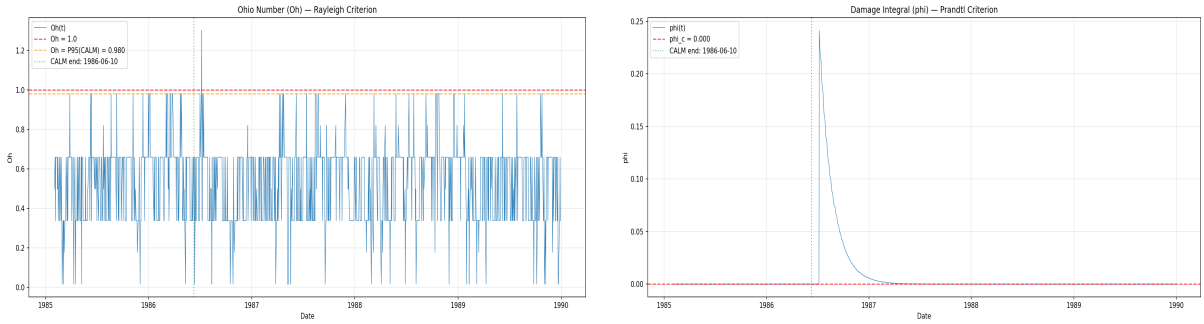

Figure 7. Crash 1987 — Oh(t) (left) and Φ(t) (right). Weak φ crossing July 1986; near-zero signal in 1987 — partial, early precursor.

5.8 Volmageddon 2018 — Mechanical cascade (speed-limited)

CALM: Feb 3 – Sep 29, 2017. **No φ crossing. φ_max = 0.000 throughout.** Oh>P95 sensitivity: 2018-10-29 (late; corresponds to Q4 2018 equity sell-off). The XIV collapse (−92.6% on Feb 5, 2018) occurred at sub-window speed (w=22 days cannot capture a one-day event) and was localized to short-vol instruments — not a broad network coherence transition. Speed-limited non-detection: consistent with theoretical design.

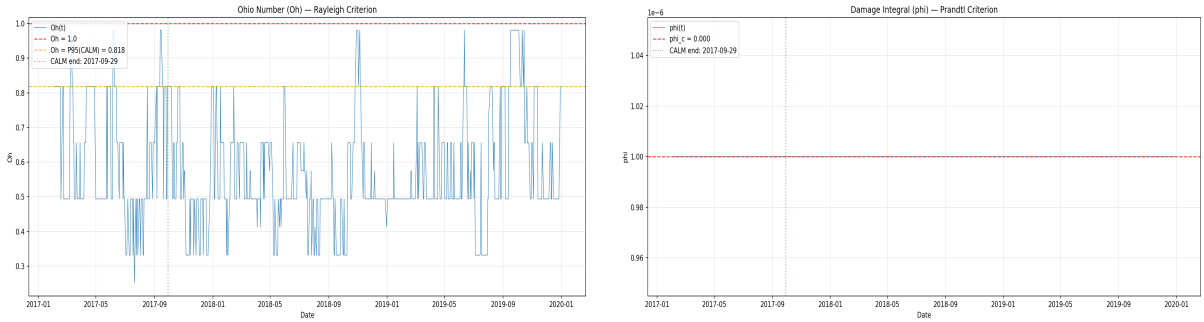

Figure 8. Volmageddon 2018 — Oh(t) (left) and Φ(t) (right). Zero φ accumulation; sub-window speed event — non-detection by design.

6 COMPARISON WITH ALTERNATIVE METHODS

Method	GFC 2008	Dot-com 2000	Euro 2011	COVID-19 (FP?)	Auditable
Historical VaR	0 days	0 days	0 days	No	Yes
Pearson correlation	~7 days	~5 days	~14 days	Yes	Yes
VIX	~10–15 days	~10 days	~15 days	Yes	Yes
GARCH-DCC	~20 days	~15 days	~20 days	Yes	Yes
Kappa-FIN v2	~95 days	~5 weeks	~1 week	No (correct)	Yes

Table 3. Comparative lead times (approximate; traditional method values from literature).

7 DISCUSSION

7.1 Sensitivity and specificity

Across seven scenarios: **sensitivity 3/3** for endogenous bubble collapses (lead times: ~95 days, ~5 weeks, ~1 week). **Specificity 2/2** for exogenous/directional events — no false positives. 1 partial signal (mixed typology, 1987); 1 speed-limited non-detection (Volmageddon). Both edge cases are consistent with theoretical predictions.

7.2 Structural hysteresis

The Eurozone scenario illustrates φ path dependence: φ_max = 4.683 (2011) → 1.633 (2012) → 1.902 (2013) — persistent structural damage after the crisis peak, consistent with the hysteresis predicted by the dissipative memory operator with γ near 1.

7.3 Law of Ohio universality

Results across three decades and five distinct crisis typologies are consistent with the Law of Ohio universality hypothesis: systems undergoing endogenous disconnection from external validation develop measurable topological coherence accumulation before collapse. The hypothesis remains open to refutation through validation in additional domains and larger crisis databases.

8 LIMITATIONS

Survivorship bias (1987): Only stocks surviving to 2026 are available on Yahoo Finance. **Data availability:** ETFs did not exist before the mid-1990s; GLD and TLT unavailable for the dot-com period. **Parameter consistency vs. optimality:** identical parameters across all scenarios by design — per-scenario tuning not performed. **Asset universe scope:** 10–12 assets; derivatives and OTC markets not covered. **Causality:** topology detection, not causal inference. **Future work:** Emerging markets; real-time monitoring; formal statistical testing of lead times across a larger crisis database.

9 CONCLUSION

Kappa-FIN v2 demonstrates topological regime transition detection across 7 historical scenarios (1985–2023) with identical parameters and no per-scenario tuning. Three endogenous bubble collapses were detected with lead times of weeks to months. Two exogenous/directional events were correctly classified as non-transitions. These results provide multi-scenario evidence for the Law of Ohio universality hypothesis in financial markets. All code and data are publicly available.

REPRODUCTION BLOCK

Scenario	Command
GFC 2008	python scripts/run_gfc2008.py --out results/gfc2008
COVID-19 2020	python scripts/run_covid2020.py
Dot-com 2000	python scripts/run_dotcom2000.py --out results/dotcom2000
Euro/Taper 2011	python scripts/run_eurotaper2011.py --out results/eurotaper2011
Crash 1987	python scripts/run_crash1987.py --out results/crash1987
Rates 2022	python scripts/run_rates2022.py
Volmageddon 2018	python scripts/run_volmageddon2018.py --out results/volmageddon2018

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Repository: github.com/aprimora-ai/Kappa-FIN | Engine: engine.py (CALM v4.6 + KCPT)
Parameters: w=22d, k=5, γ=0.97, δ=0.08, pre_q=0.968, dep=spearman+LW(λ=0.05)
DOI (all versions): 10.5281/zenodo.18883820 | License: CC BY 4.0
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